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BSIT-3

1-MONTH NETWORK FORENSICS (FEBRUARY)

WEEK 1 – Network Forensics Fundamentals & Packet Analysis

Overview

Week 1 focuses on building **the foundational knowledge required for network forensic investigations**. Network forensics involves capturing, recording, and analyzing network traffic to detect malicious activities, investigate incidents, and gather digital evidence.

During this week, the focus is on understanding **how network evidence is generated, captured, structured, and analyzed using forensic tools** such as **Wireshark, tcpdump, and tshark**.

By the end of the week, the learner should be able **to inspect packet capture (PCAP) files, identify protocols, and perform basic traffic analysis**.

1. Introduction to Network Forensics

Definition

Network forensics is a branch of digital forensics that focuses on **monitoring, capturing, recording, and analyzing network traffic** to investigate security incidents, attacks, or suspicious activities.

Unlike traditional digital forensics that analyzes data stored on devices, network forensics focuses on **data in transit across networks**.

Importance of Network Forensics

Network forensics plays an essential role in cybersecurity because it helps:

- Detect network intrusions
- Investigate cyber attacks
- Identify compromised systems
- Track attacker activity
- Provide digital evidence for legal investigations

Organizational use network forensic techniques to **reconstruct events that occurred on a network** during a security incident.

2. Role of Network Forensics in Incident Response

Network forensics is a crucial component of **incident response**, which is the process of detecting, analyzing, and responding to security incidents.

In incident response, network forensics helps investigators:

- 1) Identify the **initial attack vector**

- 2) Trace **attacker communication**
- 3) Detect **data exfiltration**
- 4) Analyze **malicious traffic patterns**
- 5) Reconstruct the **timeline of an attacker**

For example, investigator may analyze packet captures to determine whether malware communicated with a command-and-control (C2) server.

3. **Types of Network Evidence**

Network forensic investigations rely on different types of digital evidence generated by network systems.

3.1 **PCAP Files**

Packet Capture (PCAP) files store raw network traffic captured from a network interface

These files contain:

- Packet headers
- Protocol information
- Network communication details

PCAP

Files allow investigators to reconstruct **exact network communication** that occurred during an attack.

3.2 **Logs**

Network logs record activities generated by systems such as:

- Firewalls
- Routers
- Servers
- Intrusion Detection System (IDS)

Examples of logs include:

- Firewall
- Web server logs
- DNS logs
- Authentication

Logs provide **historical record of network activity**.

3.3 **NetFlow Data**

NetFlow is a network protocol used for collecting metadata about network traffic. Instead of storing full packets, NetFlows records:

- Source IP address
- Destination IP address
- Ports used

- Protocol
- Number of packets
- Duration of communication

NetFlow data is useful for identifying **traffic patterns and anomalies**

4. Evidence Handling and Chain of Custody

When conducting forensic investigations, evidence must be handled carefully to maintain its **integrity and admissibility in legal proceedings**.

Chain of Custody

Chain of custody refers to **the documented process of collecting, transferring, analyzing, and storing evidence**.

It ensures that:

- Evidence is not tampered with
- Evidence is properly documented
- Investigators maintain accountability

Typical chain-of-custody documentation includes:

- Evidence description
- Date and time of collection
- Investigator name
- Storage location
- Transfer records

Maintaining a proper chain of custody is critical in **legal and corporate investigations**.

5. OSI and TCP/IP Models (Forensics Perspective)

Understanding network models helps forensic investigators determine **where data is generated and how it travels through the network**.

OSI MODEL

The OSI model consists of **seven layers**:

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

Each layer performs a specific role in network communication.

TCP/IP MODEL

The TCP/IP model simplifies networking into **four layers**:

- I. Network Interface
- II. Internet
- III. Transport
- IV. Application

From a forensic perspective, these layers help investigators understand **which protocol or device may contain relevant evidence**.

6. *Packet Structure and Headers*

Network packets consist of two main parts:

Header

The header contains metadata about the packet, such as:

- Source address
- Destination address
- Protocol
- Packet length

Payload

The payload contains the **actual transmitted data**.

Packet headers provide valuable information for forensic analysis because they reveal **how devices communicate within a network**.

7. *Core Network Protocols*

Understanding common network protocols helps investigator identify normal and abnormal traffic.

Ethernet

- Ethernet is the standard protocol used in **local area networks (LANs)**.
- It defines how devices communicate at the **data link layer**.

Ethernet frames contain:

- Source MAC address
- Destination MAC address
- Type field
- Payload

ARP (Address Resolution Protocol)

- ARP is used to map an **IP address to a MAC address** within a local network.

Example:

If a device knows the IP address but not the MAC address, it sends an **ARP request** asking:

- “Who has this IP address?”
- The device with that IP responds with its MAC address.
- ARP traffic is often analyzed to detect **ARP spoofing attacks**.

IP (INTERNET PROTOCOL)

IP is responsible for **routing packets across networks**.

It includes:

- Source IP address
- Destination IP address
- Time to Live (TTL)
- Protocol type

Two main versions exist:

- IPv4
- IPv6

TCP vs UDP

TCP (Transmission Control Protocol)

TCP is a **connection-oriented protocol** that ensures reliable communication.

Key features:

- Three-way handshake
- Packet acknowledgement
- Error correction

TCP is commonly used in:

- HTTP
- HTTPS
- FTP
- Email protocols

UDP (User Datagram Protocol)

UDP is a **connectionless protocol**.

Characteristics:

- Faster than TCP
- No guarantee of delivery
- Minimal overhead

UDP is commonly used for:

- Video streaming
- Online gaming
- DNS queries

ICMP (Internet Control Message Protocol)

ICMP is used for **network diagnostics and error messages**.

Examples include:

- Ping request
- Destination unreachable messages
- Network troubleshooting

ICMP traffic may also be used by attackers for network reconnaissance

8. Traffic Capture Basics

Traffic capture is the process of **recording network packets for analysis**.
Captured data is typically stored in **PCAP format**.

Investigator may capture traffic using:

- Network taps
- Port mirroring
- Packet capture tools

Traffic capture allows analyst to **observe real-time or historical network communication**.

9. **Network Forensics Tools**

Several tools are commonly used for analyzing network traffic.

Wireshark

Wireshark is one of the most widely used network forensic tools.

It allows investigators to:

- Capture network packets
- Analyze protocol details
- Filter traffic
- Reconstruct sessions

Wireshark provides a **graphical interface for deep packet inspection**

Tcpdump

Tcpdump is a command-line tool used for capturing and analyzing network packets.

Example command:

tcpdump -i eth0

This command captures packets from the network interface eth0.

Tshark

Tshark is the command-line version of Wireshark.

It allows automated packet analysis and scripting.

Example command:

Tshark -r capture.pcap

This command reads packets from a PCAP file.

10. **Practical Exercises**

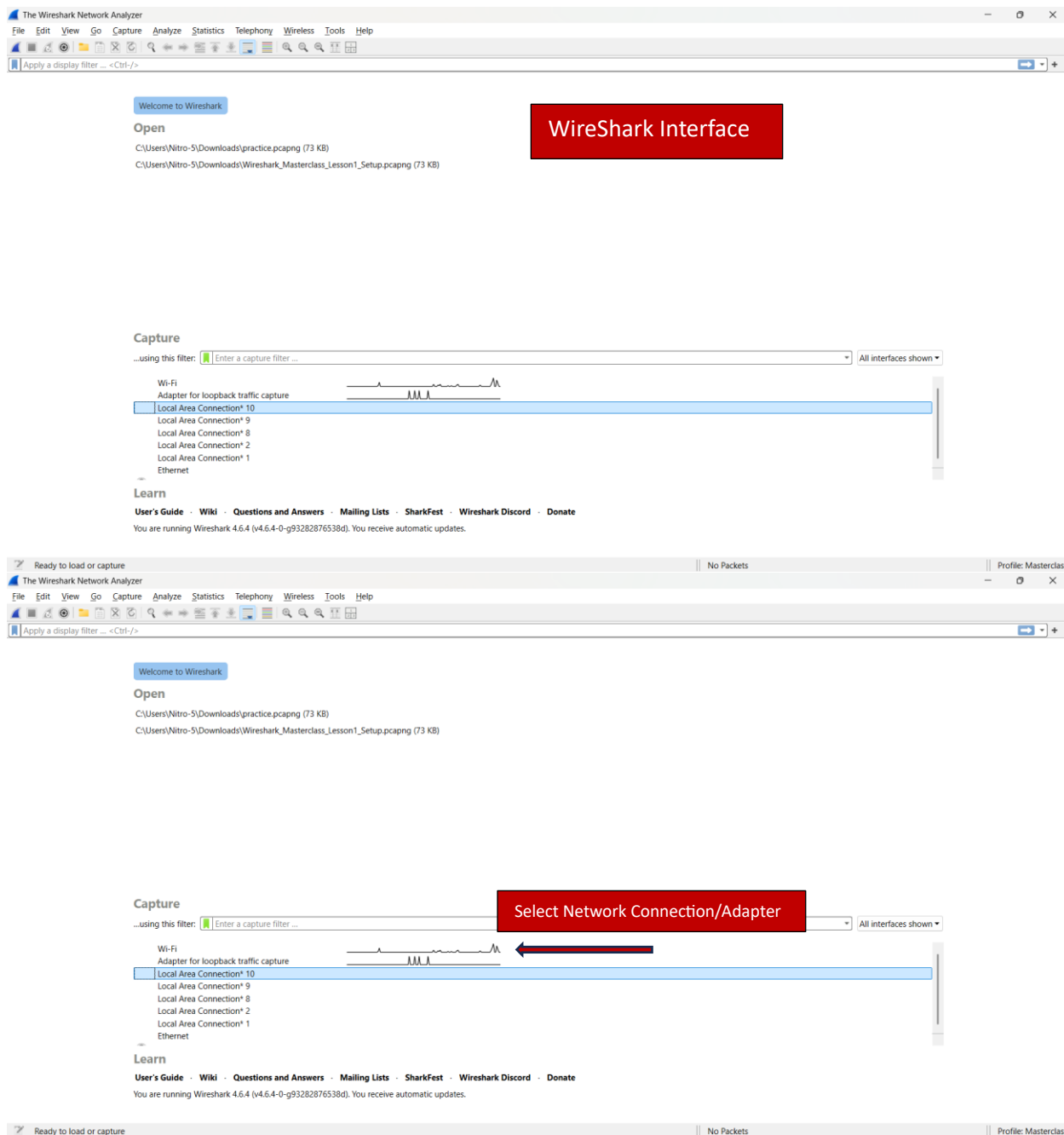
To reinforce the theoretical knowledge, several practical exercises were performed.

Opening and Inspecting PCAP Files

Steps performed:

- I. Launch Wireshark.
- II. Open a PCAP file.
- III. Observe packet summaries.
- IV. Identify source and destination addresses.

This allows analyst to examine raw network traffic.



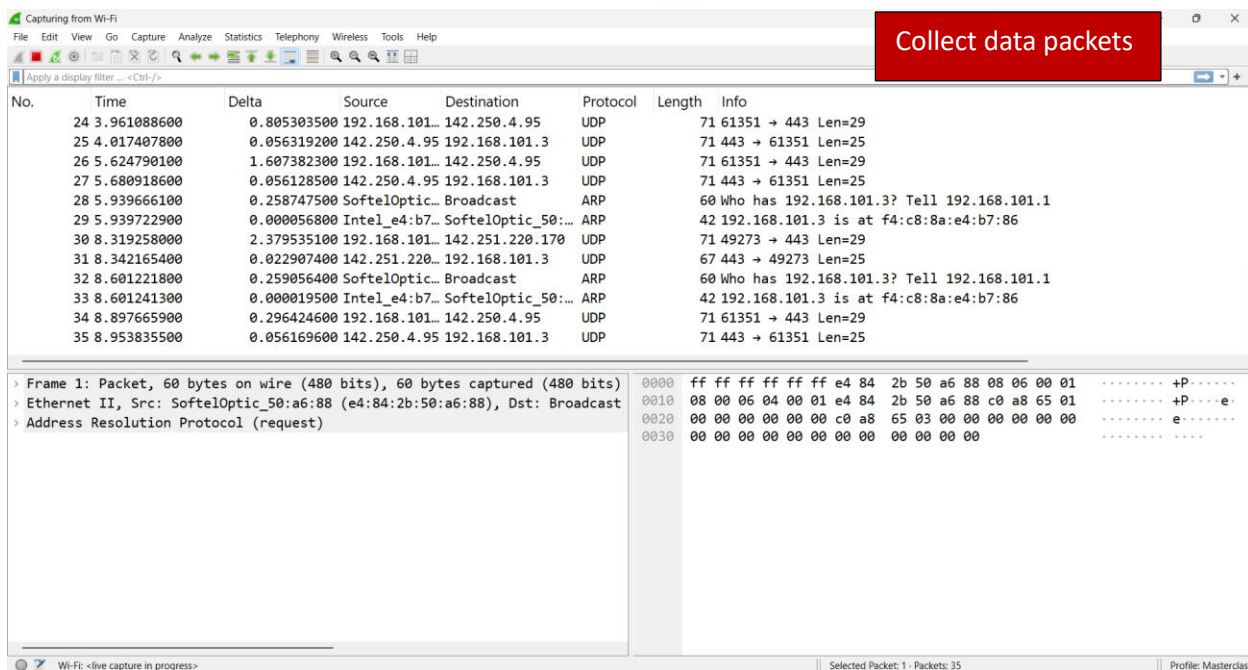
Identifying Protocols and Packet Fields

Using Wireshark, packets were inspected to identify protocols such as:

- ARP
- TCP
- UDP
- ICMP

Each packet header reveals important information such as:

- Source IP
- Destination IP
- Port numbers
- Flags



Applying Wireshark Display Filters

Display filters allow analysts to focus on specific traffic types.

Examples:

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Check packets that are using DNS server

dns

No.	Time	Delta	Source	Destination	Protocol	Length	Info
45	13.499961800		192.168.101...	192.168.101.1	DNS	91	Standard query 0xf0e3 HTTPS signaler-pa.clients6.google.com
46	13.500117300	0.000155500	192.168.101...	192.168.101.1	DNS	91	Standard query 0x02ce A signaler-pa.clients6.google.com
47	13.506071000	0.005953700	192.168.101...	192.168.101.3	DNS	151	Standard query response 0xf0e3 HTTPS signaler-pa.clients6.google.com
48	13.506071000	0.000000000	192.168.101...	192.168.101.3	DNS	107	Standard query response 0x02ce A signaler-pa.clients6.google.com
171	27.704061400	14.197990400	192.168.101...	192.168.101.1	DNS	75	Standard query 0x2cd0 HTTPS ssl.gstatic.com
172	27.704352500	0.000291100	192.168.101...	192.168.101.1	DNS	75	Standard query 0x3979 A ssl.gstatic.com
173	27.719068500	0.014716000	192.168.101...	192.168.101.3	DNS	135	Standard query response 0x2cd0 HTTPS ssl.gstatic.com SOA ns1.goog
174	27.719068500	0.000000000	192.168.101...	192.168.101.3	DNS	91	Standard query response 0x3979 A ssl.gstatic.com A 64.233.188.94
209	30.894311700	3.175243200	192.168.101...	192.168.101.1	DNS	77	Standard query 0x5c00 HTTPS gemini.google.com
210	30.895913200	0.001601500	192.168.101...	192.168.101.1	DNS	77	Standard query 0x0c7a A gemini.google.com
211	30.901408100	0.005494900	192.168.101...	192.168.101.3	DNS	93	Standard query response 0x0c7a A gemini.google.com A 142.250.199.1
215	30.941190400	0.039782300	192.168.101...	192.168.101.3	DNS	127	Standard query response 0x5c00 HTTPS gemini.google.com SOA ns1.go

> Frame 45: Packet, 91 bytes on wire (728 bits), 91 bytes captured (728 bits) on interface 0

> Ethernet II, Src: Intel_e4:b7:86 (f4:c8:8a:e4:b7:86), Dst: SoftelOptic_50:a4:00:00:00:00

> Internet Protocol Version 4, Src: 192.168.101.3, Dst: 192.168.101.1

> User Datagram Protocol, Src Port: 60391, Dst Port: 53

> Domain Name System (query)

Domain Name System: Protocol

Selected Packet: 45 - Packets: 3202 - Displayed: 120 (3.7%)

Profile: Masterclass

show only DNS traffic:

-dns

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Check packets that are using UDP

udp

No.	Time	Delta	Source	Destination	Protocol	Length	Info
534	803974000	-534.803974000	192.168.101...	142.251.220.206	UDP	71	52509 → 443 Len=29
534	778412300	0.025561700	142.251.220...	192.168.101.3	UDP	68	443 → 52509 Len=26
6	-534.395950600	0.382461700	142.250.4.95	192.168.101.3	UDP	124	443 → 61351 Len=78
7	-534.377758200	0.018192400	192.168.101...	142.250.4.95	UDP	75	61351 → 443 Len=33
8	-534.186739500	0.191018700	192.168.101...	142.250.4.95	UDP	71	61351 → 443 Len=29
9	-534.132249600	0.054489900	142.250.4.95	192.168.101.3	UDP	71	443 → 61351 Len=25
10	-533.915323000	0.216926600	192.168.101...	142.250.4.95	UDP	71	61351 → 443 Len=29
11	-533.860026000	0.055297000	142.250.4.95	192.168.101.3	UDP	71	443 → 61351 Len=25
12	-533.644499000	0.215527000	192.168.101...	142.250.4.95	UDP	71	61351 → 443 Len=29
13	-533.590338000	0.054161000	142.250.4.95	192.168.101.3	UDP	71	443 → 61351 Len=25
14	-533.376236000	0.214102000	192.168.101...	142.250.4.95	UDP	71	61351 → 443 Len=29
15	-533.321879400	0.054356600	142.250.4.95	192.168.101.3	UDP	71	443 → 61351 Len=25

> Frame 4: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface 0

> Ethernet II, Src: Intel_e4:b7:86 (f4:c8:8a:e4:b7:86), Dst: SoftelOptic_50:a4:00:00:00:00

> Internet Protocol Version 4, Src: 192.168.101.3, Dst: 142.251.220.206

> User Datagram Protocol, Src Port: 52509, Dst Port: 443

> Data (29 bytes)

User Datagram Protocol: Protocol

Selected Packet: 4 - Packets: 5980 - Displayed: 4421 (73.9%)

Profile: Masterclass

show only UDP traffic:

-udp

Capture Packets from Any Specific Website

Example: Wikipidea

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Nitro-5>ping www.wikipedia.org

Pinging dyna.wikimedia.org [103.102.166.224] with 32 bytes of data:
Reply from 103.102.166.224: bytes=32 time=41ms TTL=55
Reply from 103.102.166.224: bytes=32 time=42ms TTL=55
Reply from 103.102.166.224: bytes=32 time=41ms TTL=55
Reply from 103.102.166.224: bytes=32 time=40ms TTL=55

Ping statistics for 103.102.166.224:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 40ms, Maximum = 42ms, Average = 41ms
```

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Nitro-5>ping www.wikipedia.org

Pinging dyna.wikimedia.org [103.102.166.224] with 32 bytes of data:
Reply from 103.102.166.224: bytes=32 time=41ms TTL=55
Reply from 103.102.166.224: bytes=32 time=42ms TTL=55
Reply from 103.102.166.224: bytes=32 time=41ms TTL=55
Reply from 103.102.166.224: bytes=32 time=40ms TTL=55

Ping statistics for 103.102.166.224:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 40ms, Maximum = 42ms, Average = 41ms
```

Packet data from the website
WIKIPEDIA

Frame 20117: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
 Ethernet II, Src: Intel_e4:b7:86 (f4:c8:8a:e4:b7:86), Dst: SoftelOptic_50:a6:e0 (08:00:00:00:00:00)
 Internet Protocol Version 4, Src: 192.168.101.3, Dst: 103.102.166.224
 Transmission Control Protocol, Src Port: 64208, Dst Port: 443, Seq: 0, Len: 0

If you want to see those packets that are using only TCP PROTOCOL in the WIKIPEDIA
 "tcp and ip.addr == 103.102.166.224"

Frame 20117: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
 Ethernet II, Src: Intel_e4:b7:86 (f4:c8:8a:e4:b7:86), Dst: SoftelOptic_50:a6:e0 (08:00:00:00:00:00)
 Internet Protocol Version 4, Src: 192.168.101.3, Dst: 103.102.166.224
 Transmission Control Protocol, Src Port: 64208, Dst Port: 443, Seq: 0, Len: 0

Filters significantly improve the efficiency of forensics analysis.

Week 1 Summary

Week 1 introduced the fundamental concepts of network forensics, including network evidence types, packet structure, core protocols, and forensic tools.

Through theoretical learning and practical exercises, the learner developed the ability to analyze packet captures, apply traffic filters, and reconstruct network communications. These foundational skills are essential for conducting effective network forensic investigations and will serve as the basis for more advanced topics in the following weeks.